AI PRINCIPLES AND PRACTICE · WEEK 9

Week 9 Summary

Adapt, do not rebuild: everything to keep, in plain English

IN PLAIN ENGLISH

How to use this summary. The ten points below follow the lecture in order. Read them (about 15 minutes), try the ten questions without looking (about 15 minutes), then mark them. Re-read only the points you got wrong.

KEY IDEA

If you remember one thing. Don’t start from zero: reuse a model that already knows a lot, change it only as much as the job needs, teach it with data rather than labels, and count what it costs.

Part 1 The ten points

1.1 Reusing beats starting again, when labels are few

In the lecture's example, a network trained on 900 clean digits was reused for digits from a noisy scanner. With 5 labels per digit, a new network trained from scratch got 55% right; the reused one, 71% or more. With 1 label per digit: 62% against 34%. With 40 per digit, scratch nearly catches up. **Transfer learning** is worth most when labels are scarce.

1.2 Foundation models

One big model, trained once on a huge amount of data at huge cost, then reused for many jobs. Sizes: ResNet-50 has 25.6 million numbers, BERT-base 110 million, GPT-3 175 billion. Training such models from scratch for one job is now the exception.

1.3 Self-supervised learning: fill in the blanks

Foundation models learn without human labels by predicting hidden parts of the data from the rest: the next word, a masked word, a hidden image patch. The loss is the average surprise at the hidden pieces, L = −average log p(hidden | visible): Week 8′s cross-entropy again.

1.4 The transfer spectrum

From least to most change: **zero-shot** (use it as it is), **prompting** (change the input, not the model), **linear probe** (train only a new final layer), **LoRA** (train thin strips added to frozen tables), **fine-tuning** (train everything a little). More change needs more labels, memory and compute.

1.5 Zero-shot and prompting

Used as it is, the backbone scored 98% on clean digits but 71% on the noisy scanner: same job, new conditions. Prompting gives a language model instructions and perhaps a few examples in its input; no number in the model changes.

1.6 Linear probe, fine-tuning and LoRA

Probe: $\hat{y} = \text{softmax}(W_{\text{head}} \cdot f(x))$, 330 numbers trained, 71%. Fine-tuning: $\theta \leftarrow \theta - \eta \nabla L$ starting from the pretrained dials, all 6,570 numbers, 74%. LoRA: $W' = W + BA$, 1,226 numbers at rank 4, 73%. For a 768 × 768 table at rank 8, LoRA learns 12,288 numbers instead of 589,824; for GPT-3 it cut the numbers to train by about 10,000 times.

1.7 Augmentation encodes invariances

Augmentation adds changed copies of examples with the same labels, teaching the model what does not change the answer: f(T(x)) = f(x). Adding noisy copies of the 900 clean digits raised accuracy on the noisy scanner from 71% to 82%, with no new labels.

1.8 The wrong augmentation teaches the wrong lesson

Upside-down copies dropped accuracy on the noisy scanner to 55%, because an upside-down 6 is a 9. Only use changes that truly keep the answer.

1.9 Synthetic data: useful, with care

Data made by a model or simulator can fill gaps, protect privacy and cover rare cases, but a model trained only on its own output loses variety (**model collapse**): in the lecture's example the spread fell from 0.98 to 0.42 over 60 generations. Check synthetic data against real data.

1.10 Count the cost

Compute ≈ 6 × numbers in the model × examples ($C \approx 6ND$): GPT-3 about 3×10^{23} calculations, about a hundred trillion times our backbone's. Energy = power × time; CO_2 = energy × the grid's carbon

per kWh: 300 W for 10 hours is 3 kWh, about 0.9 kg CO_2 at 0.3 kg per kWh. GPT-3′s training was estimated at about 1,287 MWh and 552 tonnes of CO_2. Choose models by cost as well as accuracy.

Part 2 Ten questions

1. When does reusing a pretrained model help most?

A. when labels are plentiful
B. when labels are scarce
C. only for text
D. only when the job is identical

2. How do foundation models usually learn from unlabelled data?

A. by clustering
B. by predicting hidden parts of the data
C. by copying human labels
D. by random search

3. Which changes no number in the model?

A. linear probe
B. LoRA
C. prompting
D. fine-tuning

4. In a linear probe, what learns?

A. every layer
B. only a new final layer
C. thin strips added to every table
D. nothing

5. LoRA on a 1,000 × 1,000 table at rank 8 learns how many numbers?

A. 8,000
B. 16,000
C. 1,000,000
D. 64

6. What does augmentation with noisy copies teach a digit model?

A. to remember noise
B. that noise does not change the digit
C. to predict noise
D. nothing new

7. Why can upside-down copies hurt a digit model?

A. they are too dark
B. an upside-down 6 is a 9
C. they need more labels
D. they slow training

8. What is a known risk of training models only on their own output?

A. they get bigger
B. they lose variety over generations
C. they need more labels
D. nothing

9. Roughly how many calculations to train a 2-billion-number model on 10 billion words?

A. 2×10^{19}
B. 1.2×10^{20}
C. 6×10^{10}
D. 2×10^{10}

10. A 500 W card runs for 2 hours, at 0.3 kg CO_2 per kWh. How much CO_2?

A. 0.3 kg
B. 3 kg
C. 300 kg
D. 0.03 kg

Part 3 Answers

1. **B.** With 1 label per digit, reuse scored 62% against 34% from scratch. *Point 1.*
2. **B.** Self-supervised learning predicts hidden words or patches from the rest. *Point 3.*
3. **C.** Prompting changes the input only. *Point 5.*
4. **B.** The body is frozen; a new final layer learns. *Point 6.*
5. **B.** 1,000 × 8 + 8 × 1,000 = 16,000. *Point 6.*
6. **B.** Same label, different noise: the noise does not matter. *Point 7.*
7. **B.** It teaches a false invariance. *Point 8.*
8. **B.** Model collapse: variety drains away. *Point 9.*
9. **B.** $6 \times 2 \times 10^{9} \times 10^{10} = 1.2 \times 10^{20}$. *Point 10.*
10. **A.** 500 W × 2 h = 1 kWh; 1 × 0.3 = 0.3 kg. *Point 10.*